First project (Linear Regression (Coding))

→ Medical Cost Personal Dataset

Step 1 : import the required libraries

→ Read the data into csv dataframe

→ Analyze the data.

Step 2 : Feature engineering

→ Remove unnecessary columns

→ Convert the categorical data into numerical 0's & 1's.

Step 3 : Visualize the data (using seaborn, matplotlib)

→ check insights

→ derive new columns based on the visualization

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Step 4: Train test split data

→ Convert the whole into 80 – 20

→ 80% training data, 20% testing data.

Step 5: Building & training model

→ Select the model

→ fit the data to the model.

→ model.fit(x_train, y_train)

→ model = LinearRegression()

Step 6: Predicting values

→ Predict the values for testing data (20%)

Step 7: Model Evaluation

→ Evaluation metrics are how we measure how good a model is

- Mean Absolute Error (MAE)

$$\frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|$$

y_i → actual value

$\hat{y}_i$ → predicted value

- Mean Squared Error (MSE)

$$\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

- Root Mean Squared Error

$$RMSE = \sqrt{MSE}$$

$$\sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

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→ For Regression we typically use R^2 (R-squared) & Adjusted R^2 for evaluation.

- R-Squared (R^2)

$$1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

→ (residual error)^2

→ (actual - mean)^2

→ It explains the variance value of the data, ranges lies b/w 0 & 1

→ 0 → means (It explained nothing)

→ 1 → means (perfect fit (not happens))

→ jitna zyada 1 ke pass utna acha model

7-8 (good), 8-9 (v.good), 9-above (perfect).

- Adjusted R-squared

$$1 - \frac{(1-R^2)(n-1)}{n-p-1}$$

→ n = no. of rows

→ p = no. of features

→ Why?? → typically less than R^2 square

→ If we add some unnecessary feature & R^2 value increases, we use Adjusted

→ Basically it shows if the feature is really useful or not.

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Analogy → Study hr ka data hy pehle usme $R^2 \rightarrow 0.78$ arrahahy uske baad ek feature add kare fav color abh $R^2 = 0.85$ hogya normally apne ko maloom ki feature sahi nahi hy, but apan kya karte first ka adjusted R^2 nikalte phir feature add kar phir se adjusted R^2 nikalte ab agar adjusted R^2 zyada hua feature acha hy kam hua toh feature useful nahi hy.

important note : Take the n & p value of testing data not the training data.

```
n = X_test.shape[0]
p = X_test.shape[1]
```

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